Oblique Shock Waves

Compressible Fluid Mechanics, AA 504, Autumn 2026, UW

Instructor: Dr. Breidenthal

Assignment: Homework #5

Author: Emmanuel Rodriguez

emmanueljrodriguez.com

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Location: Renton, Seattle, WA

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Problem 1. Flow Visualization

Problem Statement

Based off of the movie "Flow Visualization" [3], write one paragraph on the most interesting aspect of the movie to you.

Statement

One of the most surprising moments in *Flow Visualization* comes when the researchers widen the angle of a diffuser and the flow begins to oscillate, even though the incoming water remains steady and symmetric. The hydrogen bubbles show water near one wall slowing down and then reversing direction as it encounters rising pressure. This creates a pocket of recirculating flow that pushes the main stream away from the wall. As the main stream shifts, it changes the pressure field enough to wash out the pocket. Then the pocket forms again,

and the cycle repeats. What makes the demonstration so striking is that a simple change in geometry can turn a steady inflow into a repeating pattern of separation and recovery.

Problem 2. Oblique Shock Waves - Exact Theory

Problem Statement

A flow at M1 = 2.3 approaches a ramp of 4 degrees. The upstream static pressure is p1. Determine the pressure increase, the wave angle, and the total pressure change due to the turn using exact theory.

Analysis

[*pending*]

Problem 3. Oblique Shock Waves - Linear Theory

Problem Statement

Using linear theory, determine the pressure increase and the wave angle due to the turn in Problem (2).

Analysis

[*pending*]

References

[1] Liepmann, H. W., & Roshko, A. (2001). *Elements of Gasdynamics*. Dover Publications. (Original work published 1957)

[2] Cengel, Y. (2018). *Fluid Mechanics: Fundamentals and Applications* (4th ed.). New York: McGraw-Hill.

[3] YouTube '5. Flow Visualization'.